

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

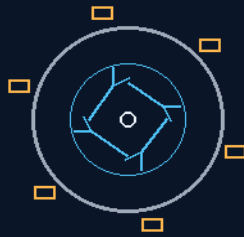
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 61

THE QUINN CENTRIFUGAL SLUICE & THERMAL-SLEEVE REFUELING CONDUITS

Gold mining in reverse — the heavies eject, the water stays
thin-film arcs hold every joint above the -40°C line



the sluice spins the grit out — the arcs keep the cold off

CRUSH TO SLUSH — CENTRIFUGAL CUT — PRE-HEAT BEFORE FLOW
TREATED AND FILTERED — HIS WORDS, THE WORLD STANDARD

no cryo valve on the hull — stubs carry the risk, charges carry the stubs

GP-IM-061

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62 of 290

VETTED: SEPARATION PASS — 'PURIFIED' ANSWERED BY HIS ORDER

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 61

PHASE 61: THE QUINN CENTRIFUGAL SLUICE & THERMAL-SLEEVE REFUELING CONDUITS — STATUS: CURRENT. Gold mining, run in reverse, in a fuel line. Phase 61 rejects high-precision, clog-prone static filters: heavy-duty mechanical crushers reduce raw ice mass to an open slush, the slush accelerates through a high-velocity centrifugal sluice, and density does the sorting — high-density rock dust, asteroid grit and glass elements forced radially outward for direct ejection back to the source harvest site, the treated and filtered low-density water-ice slurry retained at the central core axis for abrasive-free transit (the wording is his 14 August ruling: '61 is the world standard for filtered and treated water'). The conduits wear Electrotherm thin-film carbon heating arcs — virtually zero-weight semicircle elements self-aligning around the pipes, 12V/60W/100W or mains configurations, holding every joint strictly above the -40°C embrittlement threshold, sealed in Bostik Simson silyl-modified polymer. The pre-heat command fires the heating grids before any tanker fuel moves; active cryogenic valves never mount to primary hull structures — transfer routes through short sacrificial outboard stubs, and a jam triggers synchronized pneumatic ejection of all hoses, isolating both vessels instantly. Inside the pipes, disposable injection-molded plastic sleeves take the friction scoring, and worn linings are pulled, melted and re-molded in the engineering bay — a closed-loop shield.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-061, v1.0 — 22 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the sluice-and-sleeve law as seated (contents line, the four design bodies, the SMP seal-bond rulings — verbatim) — §2 why it works (density separation and thermal baselines — graded) — §3 the system in the fuel lines — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 61: **Centrifugal Sluice & Thermal-Sleeve Refueling Conduits.** Engineers high-volume propellant refueling couplers jacketed in flexible, carbon-film thermal sleeves, utilizing a low-RPM centrifugal sluice to stabilize liquid fuel density and prevent line line-lock freeze during outer depot trading runs.'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Elimination of Solar Sandblasting & Microgravity Fuel-Slosh Pump Cavitation'

REVERSE-GOLD-MINING CENTRIFUGAL SLUICE (VERBATIM)

* Matrix: Rejects high-precision, clog-prone static filters. Integrates heavy-duty mechanical crushers to reduce raw ice mass to an open slush. * Stratified Separation: Slush is accelerated through a high-velocity centrifugal sluice. Forces high-density rock dust, asteroid grit, and glass elements radially outward for direct ejection back to the source harvest site. * Resource Retention: Retains the treated and filtered low-density water ice slurry [WORDING 14 August 2026 — author's order: '61 is the world standard for filtered and treated water so it has me confused, put treated and filtered'; formerly 'purified'] inside the central core axis, ensuring abrasive-free fluid transit.

ELECTROTHERM THIN-FILM CARBON ELEMENT ARCS (VERBATIM)

* Heater Profile: Virturally zero-weight thin-film carbon heating elements formed into semicircle arcs to self-align and couple tightly around fluid conduits. * Energy Calibration: Deploys 12V (60W/100W) or 110V/240V configurations to maintain a steady thermal state strictly above the -40°C embrittlement threshold. * Silyl-Modified Polymer Sheathing: Elements are fully waterproofed and structurally sealed via Bostik Simson silyl-modified polymers, providing adhesion resilience down to the -40°C baseline.

THE PRE-HEAT COMMAND & SACRIFICIAL OUTBOARD STUBS (VERBATIM)

* Pre-Ignition Safety: Conduit heating grids activate prior to tanker fuel deployment, lifting pipe joint temperatures past brittle crack-velocity limits. * Hull Isolation Mandate: Prohibits the direct mounting of active cryogenic fluid valves onto primary hull structures, chassis docks, or skeleton frames. * Ballistic Hose Ejection: Fluid transfer routes through short, extended outboard stubs. System failures or valve jams trigger an immediate, synchronized pneumatic ejection of all hoses, isolating both vessels instantly.

DISPOSABLE INTERNAL ABRASIVE PLASTIC SLEEVES (VERBATIM)

* Core Lining Protection: Conduits are fitted with thin, disposable internal sleeves manufactured via high-impact precision plastic injection molding. * Sacrificial Wear Track: Intercepts of physical friction

scoring from moving slush, insulating primary metal walls from structural wall degradation. *

Maintenance Recyclability: Worn internal linings are pulled, melted, and re-molded within the ship's engineering bay to deliver an infinite closed-loop shield.

PHASES 62 - 64: COIL SOLAR MESH FILTERS & CENTRIFUGAL FLUID MATRICES (VERBATIM)

— the bench-proven silyl-modified-polymer seal-bond doctrine is seated INTO this phase (the thermal pipe heaters): the thermal-sleeve conduit jacket seams and coupler seals bond with SMP elastic adhesive where the joint is meant permanent. The seal's bench record, seated this day: wheel load photographed (sitting 357); water and working load passed (358); destructive-entry resistance — 'you need special machines to cut them open even after that and its hard even with them' (358); aging answered by the coach-fleet record — bonded panels, decades of high vibration (359). THE DOWNSIDE LAW, HIS WORDS ANCHORED: 'its down side is, it makes anything thinner than quarter inch steel unrepairable, you have to destroy them item to get it apart' — AFFIRMED BY MECHANISM: when the bond line's peel/shear strength exceeds the substrate's, separation tears the substrate — the bond outlives the thin part; the quarter-inch-steel line is his empirical threshold, seated as testimony. DESIGN CONSEQUENCE, SEATED: silyl-modified-polymer bonding is a build-forever decision — permanent joints only; anything meant to open gets mechanical fasteners.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE CENTRIFUGAL SLUICE, AFFIRMED: density stratification in a rotating or spiraling flow is the oldest working separation technology on Earth — the gold sluice, the hydrocyclone. Crushing to slush first and ejecting the heavies back to the harvest site is the mining step and the refining step in one motion; the vet graded the centrifugal separation PASS.

THE WORDING, RESOLVED BY THE AUTHOR: the vet's on 'purified' — needing a chemical-treatment definition — was answered 14 August on his order: '61 is the world standard for filtered and treated water so it has me confused, put treated and filtered'. The law now reads 'treated and filtered', the sluice's honest claim: mechanical separation, not chemistry.

THE THERMAL BASELINE, AFFIRMED: holding conduit joints above the -40°C embrittlement threshold with thin-film carbon arcs is standard cryogenic trace-heating, made elegant — the semicircle arcs self-align on the pipe, and the pre-heat command sequences heat before flow so no joint ever sees brittle-crack velocity. The SMP waterproofing carries the 16 August bench record: wheel load, water, working load, destructive-entry resistance — with the downside law seated: permanent joints only, anything meant to open gets mechanical fasteners.

THE SACRIFICIAL LAYERS, AFFIRMED: outboard stubs and disposable inner sleeves put the wear and the failure modes where replacement is cheap — the hull never carries an active cryo valve, a jam ejects every hose pneumatically, and the sleeve liner is a melt-and-remold loop. Failure isolation by geometry, not by hope.

§3 — THE SYSTEM IN THE FUEL LINES

- THE CRUSHERS: heavy-duty mechanical reduction of raw ice mass to open slush — no static filters to clog.
- THE SLUICE: high-velocity centrifugal separation — rock dust, asteroid grit and glass ejected radially back to the harvest site; the treated and filtered slurry retained at the core axis.
- THE HEATERS: Electrotherm thin-film carbon arcs, semicircle self-aligning, 12V (60W/100W) or 110V/240V — every joint strictly above -40°C, sealed in silyl-modified polymer.
- THE PRE-HEAT COMMAND: heating grids activate before tanker fuel deployment; no active cryogenic valve mounts to primary hull, chassis dock or skeleton frame; short outboard stubs carry the transfer; jams trigger synchronized pneumatic ejection of all hoses.
- THE SLEEVES: thin disposable injection-molded internal liners intercepting the friction scoring; worn linings pulled, melted, re-molded in the engineering bay.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE REVERSE-GOLD-MINING DOCTRINE (seated with the sluice): the oldest separation machine on Earth, pointed at ice — the heavies leave, the water stays.
- THE TREATED-AND-FILTERED LAW (his, 14 August, seated in the law text): the sluice is the world standard for filtered and treated water — mechanical honesty, no chemistry claim.
- THE PRE-HEAT DOCTRINE (seated with the command): heat before flow — a cold joint is a crack waiting for a valve to open.
- THE SMP DOWNSIDE LAW (his, 16 August, verbatim in §1): 'its down side is, it makes anything thinner than quarter inch steel unrepairable, you have to destroy them item to get it apart' — permanent joints only.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Build the crusher-sluice line: crushers to open slush, the centrifugal sluice to spec; ejection vector routed back to the harvest face.
- STEP 2 — Bench the separation: density-cut performance across grit loads; the treated-and-filtered output assayed — the world-standard line held to a number.

- STEP 3 — Wrap the arcs: thin-film carbon semicircles on every conduit; 12V and mains configurations; joint temperatures logged above -40°C under flow and stall.
- STEP 4 — Seat the SMP seams: silyl-modified polymer on the permanent joints; mechanical fasteners everywhere a joint must open — the downside law applied at the drawing stage.
- STEP 5 — Fit the stubs and ejection: outboard transfer stubs, pneumatic hose-ejection synchronized across both vessels; jam drill run until boring.
- STEP 6 — Loop the sleeves: injection-molded liners installed; the pull-melt-remold cycle timed in the engineering bay; wear ledger published.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 22 — the Babushka scissor-brace tanks (GP-IM-022): the storage the conduits feed.
- PHASE 3 — the bootstrap launch stores (GP-IM-003): the water-ice harvest the sluice refines.
- PHASE 64 — the trench-gutter stabilization (GP-IM-064): the same centrifugal family inside the tanks.
- PHASES 62-63 — the filters and the acoustic protocols (GP-IM-062/063): the shared 62-64 group this phase's masthead carries.
- PHASE 20/33 — the tethered tugs and moon depots (GP-IM-020/033): the tanker transfer routes the pre-heat command and stub ejection protect.

§8 — DO-NOT-BUILD

- NO static filter arrays — clog-prone precision is the failure mode this phase exists to delete.
- NEVER flow before heat — the pre-heat command is sequence law; a cold joint under cryo flow is a hull breach in waiting.
- NO active cryo valves on primary structure — stubs carry the risk, and the ejection charge carries the stubs.
- NEVER SMP-bond a joint that must open — the downside law: under quarter-inch steel, the bond outlives the part; mechanical fasteners only.
- NEVER say 'purified' — the law reads treated and filtered (his ruling, 14 August); chemistry claims need a chemistry plant.

§9 — OWED

THE BENCH, OWED: sluice density-cut performance across grit loads with the treated-and-filtered output

assayed; heater-arc temperature ledger under flow and stall; the sleeve wear ledger across the pull-melt-remold cycles; hose-ejection timing drill figures.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-061, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). Built 22 August 2026, fourth of the new 'next ten' shift. First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557 and 561): 'ok next 10' / 'ok next ten'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 61